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Subject Code: BMB206

Roll No:

MBA

(SEM. II) THEORY EXAMINATION 2025-26

QUANTITATIVE TECHNIQUES FOR MANAGERS

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief.

02 x 7 = 14

Q no.	Question	CO	Level
a.	In which environment, optimism criterion is used for decision making?	1	K2
b.	What is a linear programming problem? Write its mathematical form.		
c.	Differentiate between balanced and unbalanced transportation problem.		
d.	Why does transportation problem solution method not applicable to get the solution of an assignment problem?		
e.	Discuss briefly the sequencing problem with an example.		
f.	What is an M/M/1 queueing model?		
g.	Define Gantt chart with an example.		

SECTION B

2. Attempt *any three* of the following:

07 x 3 = 21

Q no.	Question	CO	Level																																							
a.	What is Operations Research? Explain the phases of Operations Research with a suitable diagram.	1	K2																																							
b.	A portfolio manager is deciding whether to invest Rs 50000 in an IT company considering his upcoming software in the market. If it is successful (probability 60 %), it will generate a revenue Rs 120000 and if it fails (40% probability), it will generate no revenue, and the investment cost is lost. Calculate the Expected Monetary Value (EMV) of this decision and determine if the project should be pursued.																																									
c.	Find the initial basic feasible solution of following transportation problem using NWC, MMM and VAM. <table><tr><td colspan="2"></td><td colspan="4">Warehouse</td><td rowspan="2">Production Capacity</td></tr><tr><td colspan="2"></td><td>W1</td><td>W2</td><td>W3</td><td>W4</td></tr><tr><td rowspan="3">Production House</td><td>A</td><td>10</td><td>12</td><td>16</td><td>13</td><td>200</td></tr><tr><td>B</td><td>16</td><td>19</td><td>15</td><td>11</td><td>250</td></tr><tr><td>C</td><td>20</td><td>21</td><td>12</td><td>10</td><td>150</td></tr><tr><td colspan="2">Storage Capacity</td><td>135</td><td>150</td><td>225</td><td>90</td><td></td></tr></table>			Warehouse				Production Capacity			W1	W2	W3	W4	Production House	A	10	12	16	13	200	B	16	19	15	11	250	C	20	21	12	10	150	Storage Capacity		135	150	225	90			
		Warehouse				Production Capacity																																				
		W1	W2	W3	W4																																					
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	B	16	19	15	11	250																																				
	C	20	21	12	10	150																																				
Storage Capacity		135	150	225	90																																					
d.	Discuss the Johnson algorithm to find the solution of N jobs through two machine.																																									
e.	What do you mean by critical path analysis? Explain the different types of slack (float) of an activity and event.																																									



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SECTION C

3. Attempt any one part of the following:

07 x 1 = 07

a.	Discuss the roll of decision tree in multi-period decision making process. Explain the decision tree and its components with suitable examples.	1	K2																									
b.	The following matrix gives the payoff (in Rs) of different alternatives A₁, A₂ and A₃ against events N₁,N₂, N₃ and N₄. <table><tr><td>Alternatives</td><td colspan="4">Events</td></tr><tr><td></td><td>N₁</td><td>N₂</td><td>N₃</td><td>N₄</td></tr><tr><td>A₁</td><td>3000</td><td>-100</td><td>5000</td><td>15000</td></tr><tr><td>A₂</td><td>19000</td><td>5000</td><td>400</td><td>0</td></tr><tr><td>A₃</td><td>20000</td><td>15000</td><td>-2000</td><td>1000</td></tr></table> Solve the above problem using the following pessimistic, optimistic, equal probability, regret and Hurwicz criterion. The degree of optimism being 0.7.	Alternatives	Events					N ₁	N ₂	N ₃	N ₄	A ₁	3000	-100	5000	15000	A ₂	19000	5000	400	0	A ₃	20000	15000	-2000	1000		
Alternatives	Events																											
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A ₁	3000	-100	5000	15000																								
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A ₃	20000	15000	-2000	1000																								

4. Attempt any one part of the following:

07 x 1 = 07

a.	What is the advantages and limitation of linear programming models? Explain the mathematical structure of linear programming problem also.	1	K2
b.	An electronic item manufacturing company produces two types of product A and B, and earn a profit of Rs. 40 and Rs 30 respectively. A single unit production of both item required processing by three operation whose monthly availability are 500, 700 and 300 hours respectively. Item a requires processing time of 4 hours , 5 hours and 2 hours of three operations while item requires 6 hours, 4 hours and 3 hours respectively for the same. Formulate the problem as LPP and find the optimal number of quantity of both products to maximize the profit of company.		

5. Attempt any one part of the following:

07 x 1 = 07

a.	Write the steps to find the optimal solution of transportation problem using MODI method.	1	K2																															
b.	Apply Hungarian method to maximize the profit for the given assignment problem: <table><tr><td></td><td colspan="4">Jobs</td></tr><tr><td></td><td>A</td><td>B</td><td>C</td><td>D</td></tr><tr><td rowspan="4">Person</td><td>1</td><td>42</td><td>24</td><td>33</td><td>39</td></tr><tr><td>2</td><td>37</td><td>22</td><td>46</td><td>24</td></tr><tr><td>3</td><td>12</td><td>31</td><td>32</td><td>35</td></tr><tr><td>4</td><td>31</td><td>24</td><td>26</td><td>40</td></tr></table>		Jobs					A	B	C	D	Person	1	42	24	33	39	2	37	22	46	24	3	12	31	32	35	4	31	24	26	40		
	Jobs																																	
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6. Attempt any one part of the following:

07 x 1 = 07

a.	Discuss the importance of queuing theory to improve the performance of a system. Explain the structure of a queuing system and its components in detail.	1	K2																								
b.	A company has five jobs and three machines A, B and C to process them. The processing order of these jobs is in ABC. In the following table processing times (in hours) are given : <table><tr><td>Jobs</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Machine A</td><td>7</td><td>10</td><td>5</td><td>8</td><td>10</td></tr><tr><td>Machine B</td><td>6</td><td>7</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Machine C</td><td>4</td><td>9</td><td>5</td><td>8</td><td>5</td></tr></table> Determine a sequence for the five jobs that will minimize the elapsed time and idle time for machines.	Jobs	1	2	3	4	5	Machine A	7	10	5	8	10	Machine B	6	7	3	4	5	Machine C	4	9	5	8	5		
Jobs	1	2	3	4	5																						
Machine A	7	10	5	8	10																						
Machine B	6	7	3	4	5																						
Machine C	4	9	5	8	5																						

7. Attempt any one part of the following:

07 x 1 = 07

a.	Which techniques is used for project scheduling with uncertain activity times? Discuss it in detail with suitable example.	1	K2																																	
b.	The following maintenance job has to be performed on regular basis in a company: <table><tr><th>Task / activity</th><th>Immediate Predecessor</th><th>Time (in days)</th></tr><tr><td>A</td><td>-</td><td>14</td></tr><tr><td>B</td><td>A</td><td>22</td></tr><tr><td>C</td><td>B</td><td>10</td></tr><tr><td>D</td><td>B</td><td>16</td></tr><tr><td>E</td><td>B</td><td>12</td></tr><tr><td>F</td><td>C</td><td>10</td></tr><tr><td>G</td><td>C</td><td>6</td></tr><tr><td>H</td><td>F,G</td><td>8</td></tr><tr><td>I</td><td>D,E,H</td><td>24</td></tr><tr><td>J</td><td>I</td><td>16</td></tr></table> a. Draw the network diagram of problem. b. Identify the critical path and its duration. c. Calculate the total float and free float for each activity.	Task / activity	Immediate Predecessor	Time (in days)	A	-	14	B	A	22	C	B	10	D	B	16	E	B	12	F	C	10	G	C	6	H	F,G	8	I	D,E,H	24	J	I	16		
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